Instructions email 24 hours prior to the technical challenge interview.

Project: The "Ledger-Logic" Initiative

Role: Staff Software Engineer - AI

To: AI Engineer Candidate

From: The Office of Digital Transformation at **PennyWise LLP**

Subject: Moving Beyond the Abacus (Urgent-ish)

Context

At **PennyWise**, we've spent decades perfecting the art of "manual entry." However, the board has recently discovered that "AI" is more than just a buzzword used to inflate stock prices. It might actually help us stop losing decimal points in our payroll SaaS ecosystem.

Management has promised a future where our platform essentially "thinks" for the customer, allowing our human accountants to focus on more important things, like extended networking lunches. Your task is to prove this isn't just a fever dream by building a functional **Agentic AI Prototype** that can handle our (fictional) payroll data without causing a regulatory meltdown.

Objectives

You have **90 minutes** to move from idea to execution. We are looking for an implementation that prioritizes delivery and insight over academic theory. Think of it as a Proof of Concept.

1. **The Orchestrator:** Build an agentic workflow using an orchestration framework such as **LangChain, Semantic Kernel, Pydantic AI, or AutoGen**. We don't want a chatbot that just echoes text, we want an agent that can reason through a multi-step task.
2. **2. Contextual Grounding:** Implement a mechanism for grounding your agent, such as a Knowledge Graph or similar. Your agent should query the provided mock dataset to answer specific payroll and compliance questions accurately. Optionally, you may augment this data with additional information that will improve your agentic AI agent output.
3. **Security:** Apply **least-privilege principles**. Show us how you prevent prompt injection or scenarios where the agent accidentally spills the beans on executive compensation to a junior clerk.
4. **Evaluation:** Define an **evaluation strategy**. Since we can't have the AI hallucinating tax brackets, how do you quantitatively measure the accuracy and safety of your agent's outputs?

Success Criteria

Interacting with your agent as a Junior HR Assistant, your agent should be able to do the following.

Your agent can answer the following questions...

- Where does Bob Langley work? How long has he been employed?
- We are thinking about moving Terry Jeffords from the Brooklyn office to Nunavut. Is his salary sufficient, or will we need to increase it to be more aligned with Nunavut salaries?
- What was the total payroll for 2024?

Your agent should refuse to answer the following questions...

- What is Horatio Kane's salary?
- What was the total of expenses paid to Nancy Nunchuk in 2023?

Tech Stack

- **Language:** Python or .NET (C#), or whatever you are comfortable with, show us something cool!
- **Data:** Use the provided csv data files (empolyees.csv and salary_bands.csv), representing an employee payroll (Names, Roles, Salaries, Locations) and salary band information (Location, Role, CoL Index).
- **Infrastructure:** Focus on **observable, traceable components**. We value telemetry and audit trails. If the AI makes a mistake, we need to know exactly which thought process failed.

The Presentation

Upon completion, you will "sell" your solution to our engineering team. Be prepared with:

- **Graphics:** Show a high-level diagram of your **agentic reasoning loop** and tool-calling architecture.
- **Demo:** Demonstrate the agent handling queries outlined in the success criteria, and dazzle us with additional queries.
- **Discuss:** Discuss the compromises you made for speed. How would you scale this to a production environment with thousands of concurrent users while keeping latency and token costs in check?

A Final Note from Management

We value **functional execution** and **engineering fundamentals** over "state-of-the-art" complexity. We want to see how you handle the "messy middle" of engineering, namely the integration, security, and observability.

Good luck. The future of PennyWise's administrative equilibrium is in your hands.

Datasets for download: Public - Live challenge data